

Linear Graphs $y = mx + c$

$y = mx + c$: m is the steepness, c is where the line crosses the y -axis. Read them, draw with them, and build an equation from any two facts.



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THE TOPIC ON ONE MAP

Linear Graphs $y = mx + c$

01 Draw a line from its equation

table of values

$$y = 7 - 4x \text{ for } -2 \leq x \leq 3$$

02 Read m and c

$3y = 3x - 2$: intercept

which lines are parallel?

03 Equation from a graph or two points

line drawn on a grid

through $(1, -1)$ and $(4, 7)$

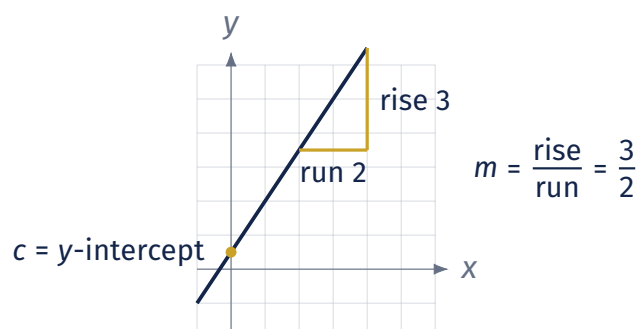
04 Parallel and perpendicular lines

parallel through $(3, 4)$

perpendicular through P

05 Lines and shapes

area of triangle OAB



01 Draw a Line from its Equation

Make a table: choose x values across the range, work out y , plot, join with a ruler. Three points — the third one checks the other two.

WHEN YOU SEE → YOU DO

“draw the graph of $y = 3x - 2$ for x from -1 to 3 ”

→ table: $x = -1, 0, 1, 2, 3$

“complete the table of values”

→ substitute each x — watch negatives

a negative gradient $y = 8 - 2x$

→ the line goes down from left to right

“for values of x from -2 to 4 ”

→ draw from the first x to the last — no further

WORKED MODEL • TABLE

Draw $y = 7 - 4x$ for $-2 \leq x \leq 3$.

1 $x = -2, 0, 3 \rightarrow y = 15, 7, -5$

table

2 plot $(-2, 15), (0, 7), (3, -5)$; join with a ruler

draw

ANSWER a straight line from $(-2, 15)$ to $(3, -5)$

× **TRAP** Joining the points “by eye” into a curve, or stopping short of the given x values.

IDEA 01

Practise: Draw a Line from its Equation

My questions first, then selected practice, then real past papers.

Q1 DR ESLAM Table

CH 3 • TOPIC 4 3 marks

Draw the graph of $y = 2x - 1$ for $-1 \leq x \leq 3$. Write your table first.

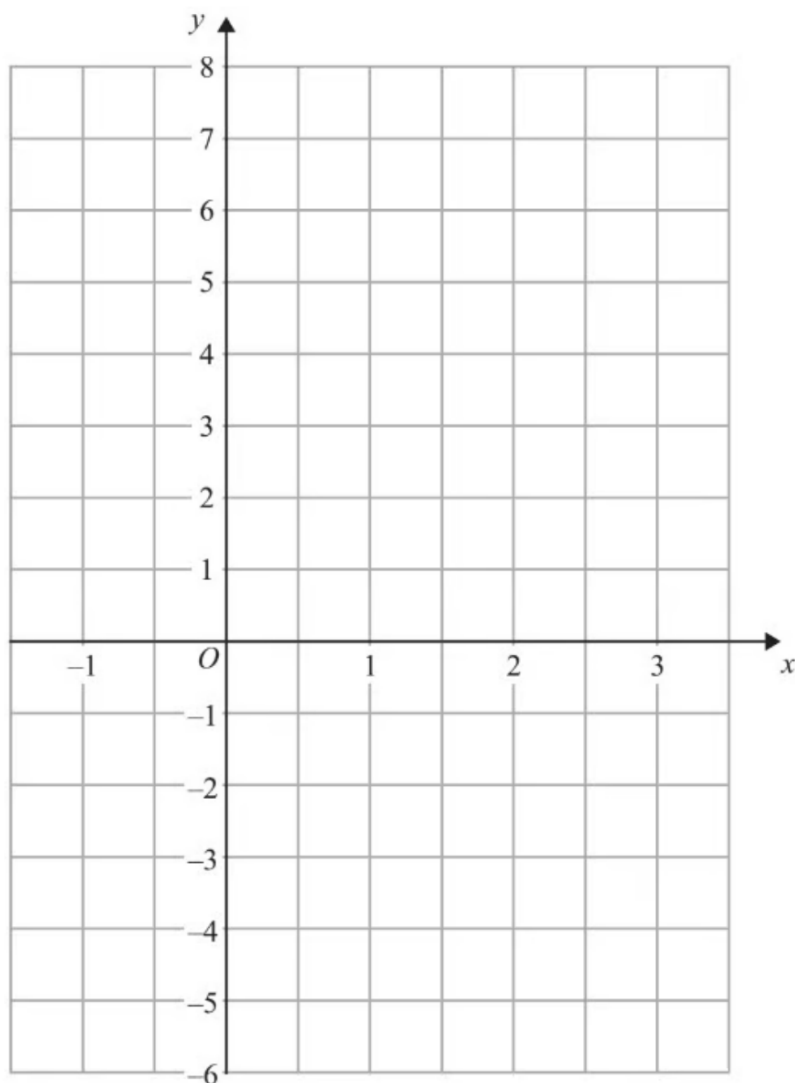
SELECTED PRACTICE

Q2**PRACTICE**

Practice · Easy

CH 3 · TOPIC 4 3 marks

- 1 On the grid, draw the graph of $y = 3x - 2$ for values of x from -1 to 3



(3 marks)

Q3**PRACTICE**

Practice · Easy

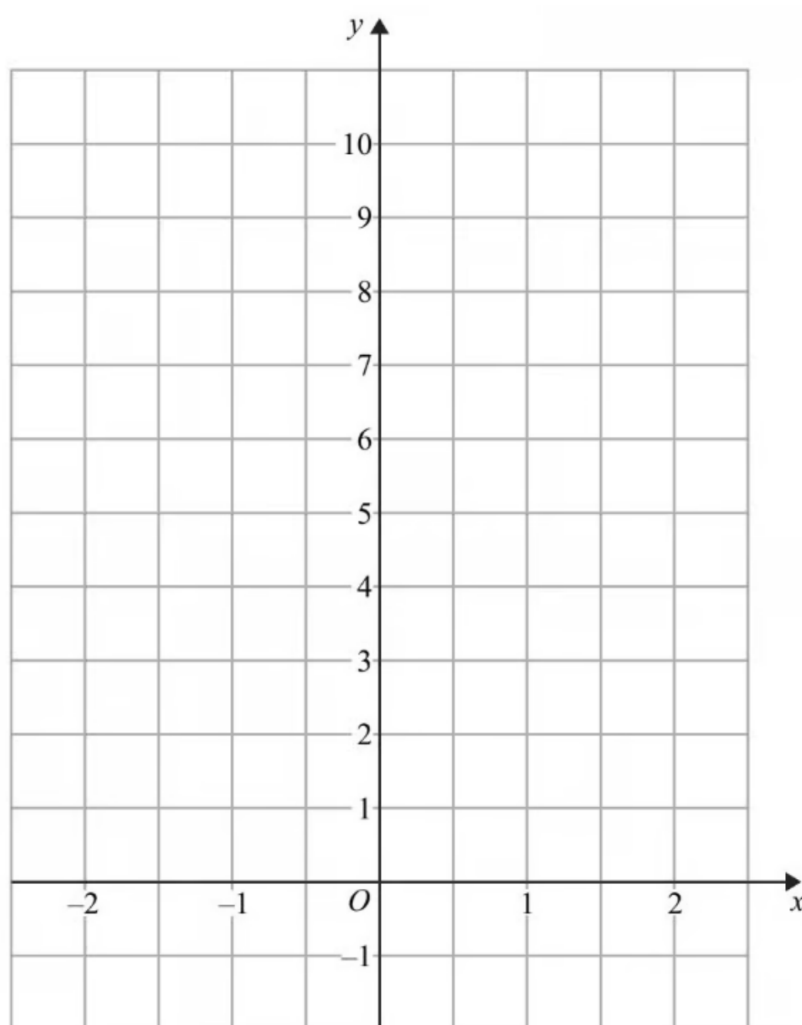
CH 3 · TOPIC 4

4 marks

4 (a) Complete the table of values for $y = 2x + 5$

x	-2	-1	0	1	2
y	1		5		

(2 marks)

(b) On the grid, draw the graph of $y = 2x + 5$ for values of x from $x = -2$ to $x = 2$ 

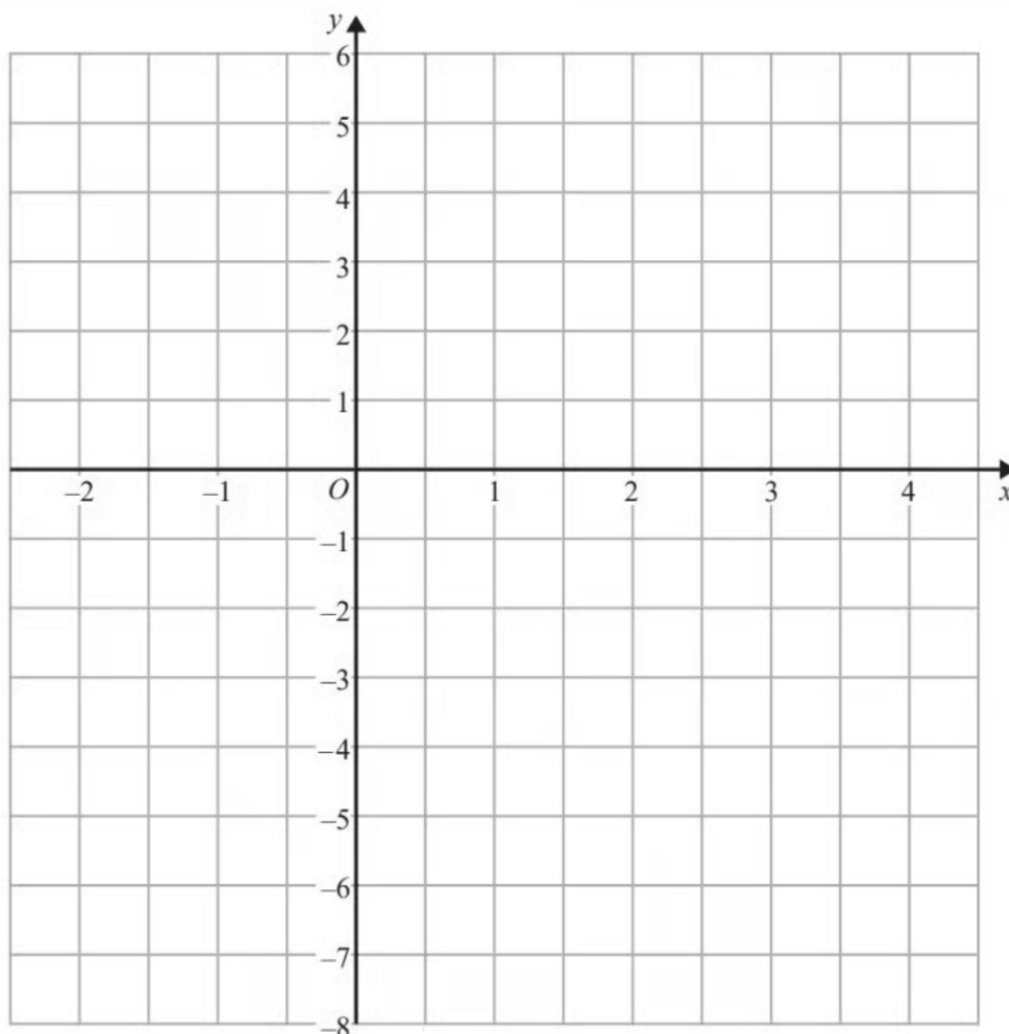
(2 marks)

Q4**PRACTICE**

Practice • Easy

CH 3 • TOPIC 4 3 marks

- 7 On the grid below, draw the graph of $y = 2x - 3$ for values of x from -2 to 4

**(3 marks)**

Q5**PRACTICE**

Practice · Medium

CH 3 · TOPIC 4**4 marks**

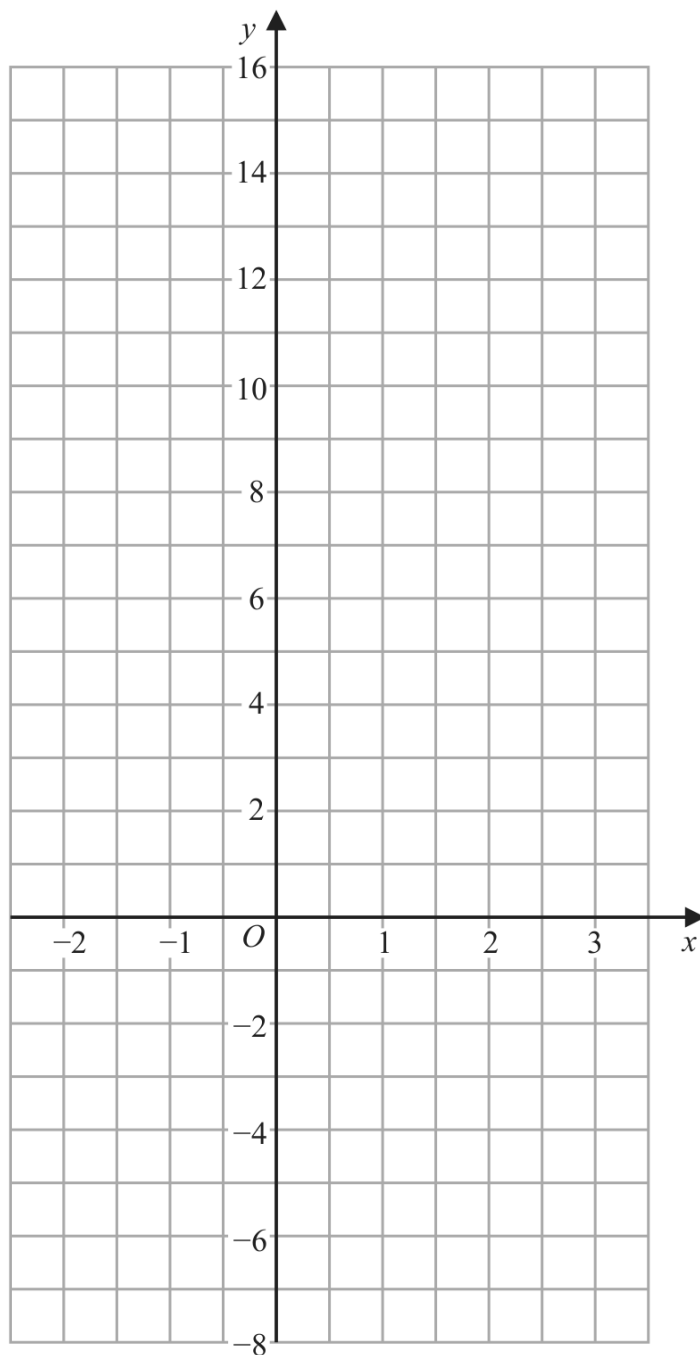
2 On the grid, draw the graph of $y = 3x + 2$ for values of x from -2 to 2

(4 marks)

PAST PAPERS

Q6 **PAST PAPER** Edexcel 4MA1 • Oct/Nov 2020 • Paper 1H • Q3**CH 3 • TOPIC 4** **3 marks**

- 3 On the grid, draw the graph of $y = 7 - 4x$ for values of x from -2 to 3



(Total for Question 3 is 3 marks)

02 Read m and c

Make y the subject. The number in front of x is the gradient; the number on its own is the y -intercept. Parallel lines have the same gradient.

WHEN YOU SEE → YOU DO

“ $2y + 7x = 10$: gradient / y -intercept”

→ $y = -\frac{7}{2}x + 5$: $m = -\frac{7}{2}$, $(0, 5)$

“crosses the x -axis”

→ put $y = 0$

“which two lines are parallel?”

→ rearrange each; same m

“show that these two lines are parallel”

→ rearrange both, compare m

“is the point above, below or on the line?”

→ substitute x , compare y

WORKED MODEL • PARALLEL

Show that $y = 3x - 2$ and $3y - 9x + 5 = 0$ are parallel.

1 $3y = 9x - 5 \Rightarrow y = 3x - \frac{5}{3}$

y the subject

ANSWER both have $m = 3$, so they are parallel

IDEA 02

Practise: Read m and c

My questions first, then selected practice.

Q7

DR ESLAM

Read

CH 3 • TOPIC 4

2 marks

Find the gradient and y-intercept of $4x + 2y = 6$.

SELECTED PRACTICE**Q8**

PRACTICE

Practice • Easy

CH 3 • TOPIC 4

1 mark

8 The straight line **L** has equation $2y + 7x = 10$

Find the coordinates of the point where **L** crosses the y-axis.

(1 mark)

Q9**PRACTICE**

Practice • Easy

CH 3 • TOPIC 4 1 mark

- 11 A line has equation $3y = 3x - 2$

Circle the coordinates of the intercept of the line with the y -axis.

$(0, 1)$	$(0, -1)$	$\left(0, \frac{2}{3}\right)$	$\left(0, -\frac{2}{3}\right)$
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(1 mark)

Q10**PRACTICE**

Practice • Medium

CH 3 • TOPIC 4 2 marks

- 3 The equation of the line L_1 is $y = 3x - 2$
 The equation of the line L_2 is $3y - 9x + 5 = 0$

Show that these two lines are parallel.

(2 marks)

Q11 PRACTICE Practice · Medium

CH 3 · TOPIC 4 1 mark

8 Here are the equations of four straight lines.

Line A $y = 2x + 4$

Line B $2y = x + 4$

Line C $2x + 2y = 4$

Line D $2x - y = 4$

Two of these lines are parallel.

Write down the two parallel lines?

(1 mark)

Q12 PRACTICE Practice · Medium

CH 3 · TOPIC 4 1 mark

12 The equation of a straight line is $3x + 2y = 24$

Circle the point where the line crosses the x -axis.

(0, 8)	(12, 0)	(0, 12)	(8, 0)
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(1 mark)

Q13**PRACTICE**

Practice · Hard

CH 3 · TOPIC 4 5 marks

13 (a) Show that the lines $y = 3x + 7$ and $2y - 6x = 8$ are parallel.

Do **not** use a graphical method.

(3 marks)

(b) Is the point $(-5, -6)$ above, below or on the line $y = 3x + 7$?

Tick **one** box.

☐ Above ☐ Below ☐ On the line

You **must** show your working.

Do **not** use a graphical method.

(2 marks)

03 Equation from a Graph or Two Points

From a graph: read c where it crosses the y -axis, find m with a big triangle (rise $\div$ run). From two points: m first, then $y - y_1 = m(x - x_1)$.

WHEN YOU SEE $\rightarrow$ YOU DO

a line drawn on a grid	$\rightarrow$ c from the y -axis; m from a big triangle
"the scales are not shown"	$\rightarrow$ use the known line to find the scale
"through $(1, -1)$ and $(4, 7)$ "	$\rightarrow$ $m = \frac{8}{3}$, then substitute
"in the form $ax + by = c$ "	$\rightarrow$ clear fractions, x and y on one side
a real-life line (height, cost)	$\rightarrow$ c = start value, m = rate

Read c Big triangle for m $y = mx + c$

WORKED MODEL • JUNE 2022 P1HR

Line L crosses the y -axis at -1 and passes through $(4, 5)$. Find $y = mx + c$.

1 $c = -1$

y-intercept

2 $m = \frac{5 - (-1)}{4 - 0} = \frac{3}{2}$

rise $\div$ run

ANSWER $y = 1.5x - 1$

IDEA 03

Practise: Equation from a Graph or Two Points

My questions first, then selected practice, then real past papers.

Q14 DR ESLAM Two points

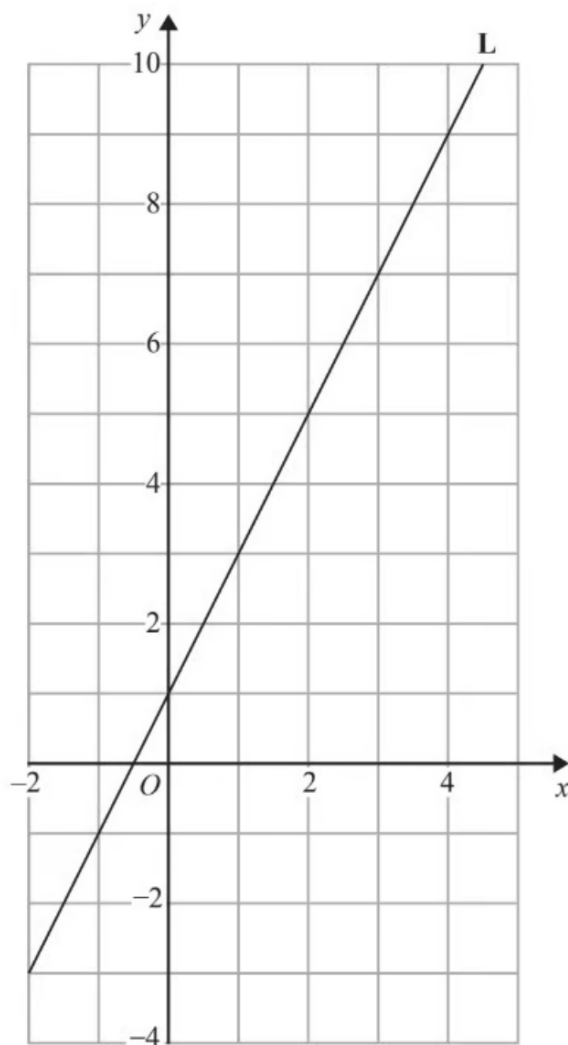
CH 3 • TOPIC 4 3 marks

Find the equation of the line through (2, 5) and (6, 13).

SELECTED PRACTICE

Q15 PRACTICE Practice · EasyCH 3 · TOPIC 4 **3 marks**

2 Line **L** is drawn on the grid below.



Find the equation for the straight line **L**.

Give your answer in the form $y = mx + c$

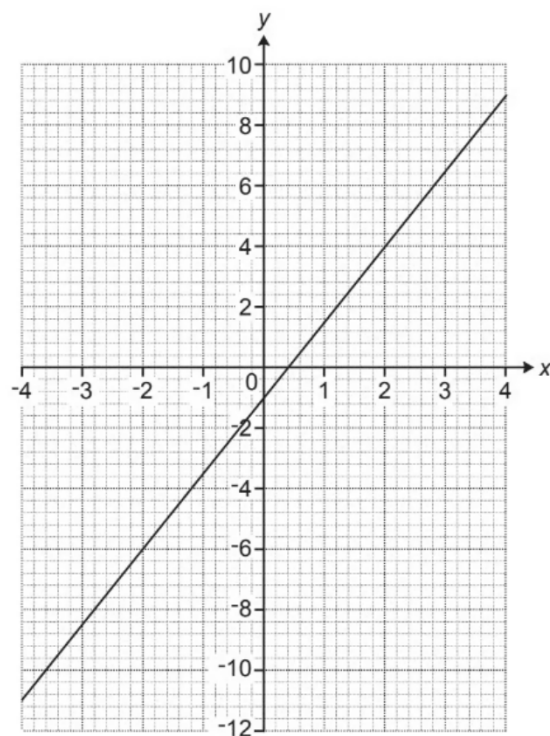
(3 marks)

Q16**PRACTICE**

Practice • Easy

CH 3 • TOPIC 4 3 marks

13 (a) This graph shows part of a straight line.



Show that the gradient of the line is 2.5.

(1 mark)

(b) Write down the equation of the line.

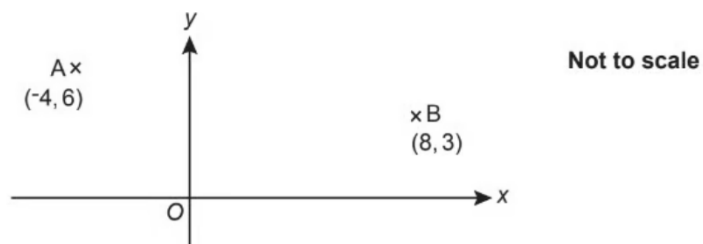
(2 marks)

Q17**PRACTICE**

Practice · Medium

CH 3 · TOPIC 4 6 marks

- 17 (a) Point A has coordinates $(-4, 6)$ and point B has coordinates $(8, 3)$.



- (i) Find the gradient of line AB.

[2]

- (ii) Find the equation of line AB.

[2]

(4 marks)

- (b) Point P has coordinates $(0, -2)$.

Write down the equation of the line parallel to line AB that passes through P.

(2 marks)

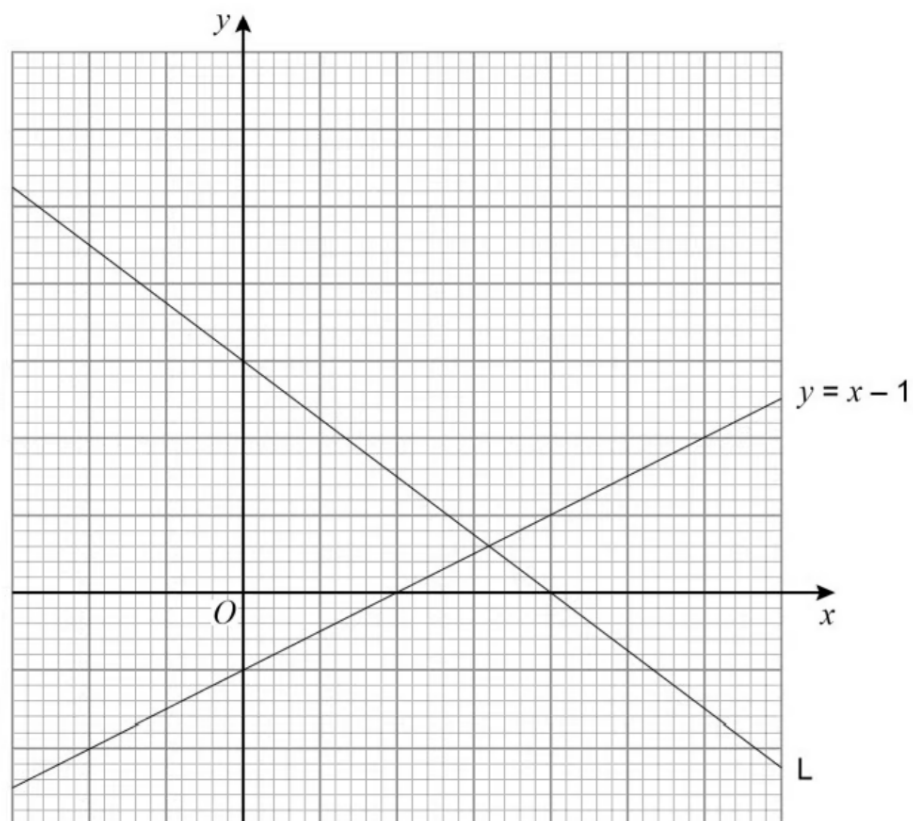
Q18**PRACTICE**

Practice • Hard

CH 3 • TOPIC 4 4 marks

12 Here is line L and the graph of $y = x - 1$

The scales of the axes are not shown.



Work out the equation of line L.

(4 marks)

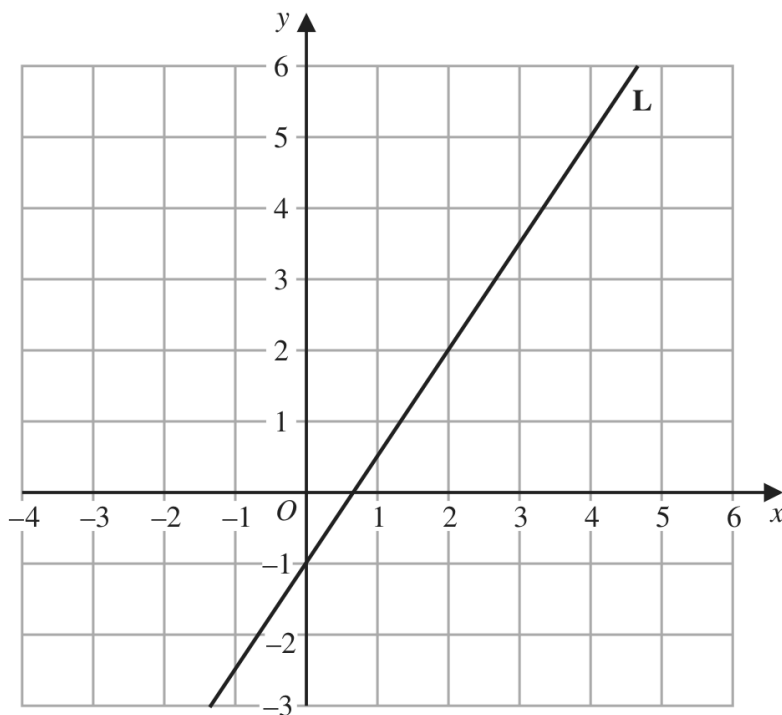
PAST PAPERS

Q19**PAST PAPER**

Edexcel 4MA1 • June 2022 • Paper 1HR • Q10

CH 3 • TOPIC 4 3 marks

10 Line **L** is drawn on the grid.



Find an equation for **L**

Give your answer in the form $y = mx + c$

(Total for Question 10 is 3 marks)

Q20**PAST PAPER**

Edexcel 4MA1 • November 2023 • Paper 2H • Q9

CH 3 • TOPIC 4 2 marks

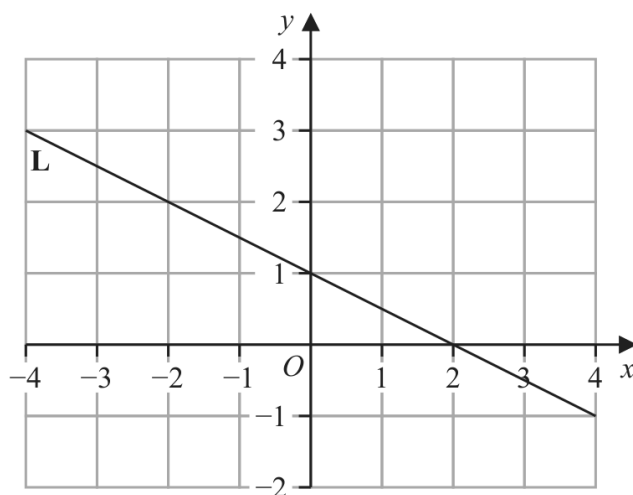
- 9 A straight line passes through the points with coordinates $(0, -3)$ and $(2, 0)$

Find an equation of the line.

.....
(Total for Question 9 is 2 marks)

Q21 PAST PAPER Edexcel 4MA1 • May 2024 • Paper 1HR • Q9

CH 3 • TOPIC 4 3 marks

9 Line **L** is drawn on the grid.Find an equation for **L**Give your answer in the form $y = mx + c$

(Total for Question 9 is 3 marks)

04 Parallel and Perpendicular Lines

Parallel: same m . Perpendicular: flip and change sign. Then use the point to find c .

WHEN YOU SEE → YOU DO

“parallel to $y = 3x + 2$, through $(3, 4)$ ”

$$\rightarrow m = 3: 4 = 9 + c$$

“perpendicular to $2x + y = 9$ ”

$$\rightarrow m = -2 \rightarrow \frac{1}{2}$$

“is Michael correct — are they perpendicular?”

$$\rightarrow m_1 \times m_2 = -1?$$

“the perpendicular bisector of PQ ”

$\rightarrow$ midpoint and perpendicular gradient

ZONE PARALLEL

same m

$$y = 3x + 2 \text{ and } y = 3x - 5$$

ZONE PERPENDICULAR

$$m_2 = -\frac{1}{m_1}$$

$$m = \frac{2}{3} \text{ and } m = -\frac{3}{2}$$

WORKED MODEL • NOVEMBER 2024 P2H

L is perpendicular to $2x + y = 9$ and passes through $(8, 11)$. Find L .

1 $y = -2x + 9: m = -2 \rightarrow m_L = \frac{1}{2}$

perpendicular

2 $11 = \frac{1}{2}(8) + c \Rightarrow c = 7$

use the point

ANSWER $y = \frac{1}{2}x + 7$

IDEA 04

Practise: Parallel and Perpendicular Lines

My questions first, then selected practice, then real past papers.

Q22 DR ESLAM Perpendicular

CH 3 · TOPIC 4 3 marks

Find the line perpendicular to $y = 3x - 1$ through $(6, 2)$.

SELECTED PRACTICE**Q23** PRACTICE Practice · Easy

CH 3 · TOPIC 4 1 mark

10 Write down an equation of a line that is parallel to the line with equation $y = 7 - 4x$

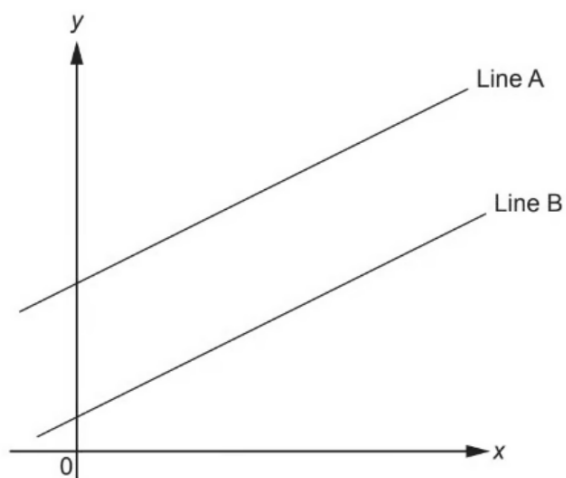
(1 mark)

Q24**PRACTICE**

Practice • Medium

CH 3 • TOPIC 4 4 marks

- 15 The graph shows two parallel lines, Line A and Line B.

**Not to scale**

Line A has equation $y = 6x + 7$.

Line B passes through the point (4, 26).

Find the equation of Line B.

(4 marks)

Q25**PRACTICE**

Practice • Hard

CH 3 • TOPIC 4 3 marks

- 3 The straight line **L** has the equation $3y = 4x + 7$

The point **A** has coordinates $(3, -5)$

Find an equation of the straight line that is perpendicular to **L** and passes through **A**.

(3 marks)**Q26****PRACTICE**

Practice • Very Hard

CH 3 • TOPIC 4 4 marks

- 11 Line A has equation $y = 4x - 1$

Line B is

perpendicular to line A

and

passes through the point $(8, 5)$

Work out the coordinates of the point where line B intersects the x -axis.

(4 marks)

PAST PAPERS

Q27**PAST PAPER**

Edexcel 4MA1 • June 2024 • Paper 2H • Q18

CH 3 • TOPIC 4 2 marks

18 The straight line **P** has equation $5y + 2x = 7$

Find the gradient of a straight line that is perpendicular to **P**

(Total for Question 18 is 2 marks)

Q28**PAST PAPER**

Edexcel 4MA1 • November 2024 • Paper 2H • Q18

CH 3 • TOPIC 4 4 marks

18 The straight line **L** is perpendicular to the straight line with equation $2x + y = 9$ and passes through the point with coordinates (8, 11)

Find an equation for **L**

Give your answer in the form $y = mx + c$

(Total for Question 18 is 4 marks)

Q29**PAST PAPER**

Edexcel 4MA1 • June 2025 • Paper 2HR • Q10

CH 3 • TOPIC 4 2 marks

- 10** A straight line, **L**, is parallel to the line with equation $y = 2 - 5x$
The line **L** passes through the point (0, 6)

Find an equation of the line **L**

.....
(Total for Question 10 is 2 marks)

05 Lines and Shapes

Find where lines cross the axes ($x = 0$ or $y = 0$) and each other (solve together). Then use the shape: right angles are perpendicular gradients.

WHEN YOU SEE → YOU DO

“area of triangle OAB ”

→ intercepts give base and height

“ l_2 passes through B and is perpendicular to l_1 ”

→ B from l_1 , then $-\frac{1}{m}$

rectangle / square / rhombus

→ sides meet at right angles; rhombus diagonals $\perp$

“find the length of PD ”

→ both points on the y -axis: subtract

WORKED MODEL • TRIANGLE AREA

$l_1: 2x - 3y + 12 = 0$ crosses the axes at A and B . $l_2 \perp l_1$ through B meets the x -axis at C . Area of ABC ?

1 $A(-6, 0), B(0, 4), m_1 = \frac{2}{3}$

intercepts

2 $l_2: y = -\frac{3}{2}x + 4 \rightarrow C\left(\frac{8}{3}, 0\right)$

perpendicular

ANSWER $\frac{1}{2} \times \frac{26}{3} \times 4 = \frac{52}{3}$

IDEA 05

Practise: Lines and Shapes

My questions first, then selected practice.

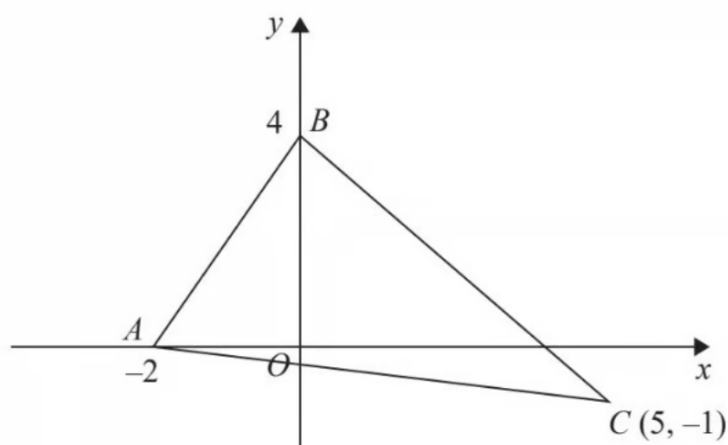
SELECTED PRACTICE**Q30**

PRACTICE

Practice • Hard

CH 3 • TOPIC 4 4 marks

4

Find an equation of the line that passes through C and is perpendicular to AB .**(4 marks)**

Q31**PRACTICE**

Practice · Hard

CH 3 • TOPIC 4**4 marks**

10 The straight line **L** passes through the points $(4, -1)$ and $(6, 4)$

The straight line **M** is perpendicular to **L** and intersects the y -axis at the point $(0, 8)$

Find the coordinates of the point where **M** intersects the x -axis.

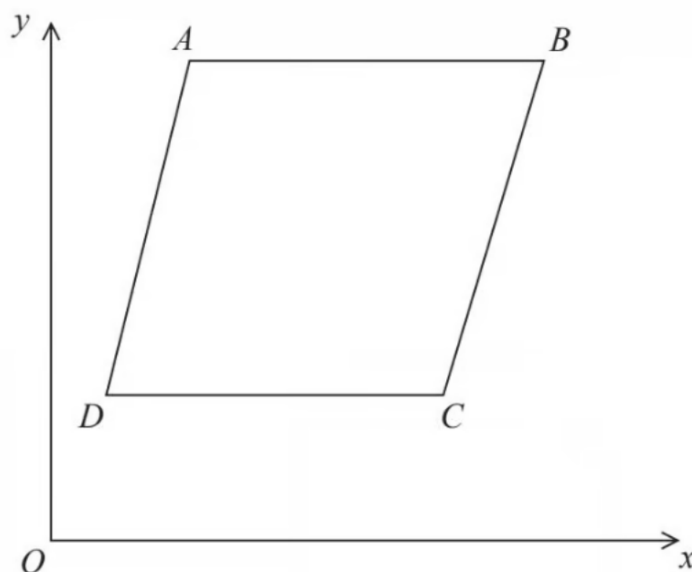
(4 marks)

Q32**PRACTICE**

Practice · Very Hard

CH 3 · TOPIC 4 4 marks

1



$ABCD$ is a rhombus.

The coordinates of A are $(5, 11)$

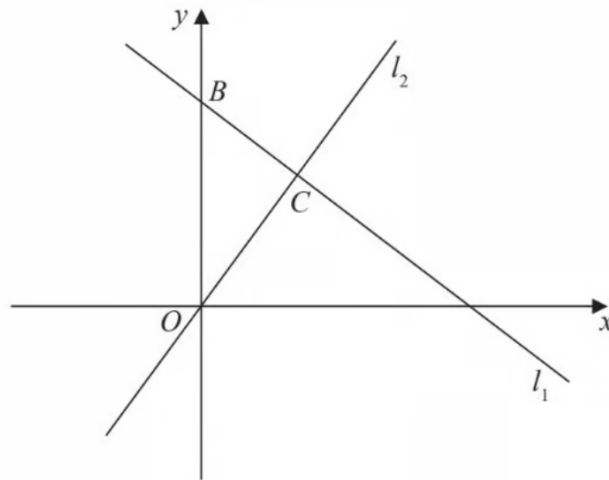
The equation of the diagonal DB is $y = \frac{1}{2}x + 6$

Find an equation of the diagonal AC .

(4 marks)

Q33**PRACTICE**

Practice • Very Hard

CH 3 • TOPIC 4 10 marks**3 (a)****Figure 2**

The line l_1 , shown in Figure 2 has equation $2x + 3y = 26$

The line l_2 passes through the origin O and is perpendicular to l_1

Find an equation for the line l_2

(4 marks)

- (b)** The line l_2 intersects the line l_1 at the point C . Line l_1 crosses the y -axis at the point B as shown in Figure 2.

Find the area of triangle OBC .

Give your answer in the form $\frac{a}{b}$, where a and b are integers to be determined.

(6 marks)

Q34**PRACTICE**

Practice • Very Hard

CH 3 • TOPIC 4 5 marks

8 ABC is an isosceles triangle with $AB = AC$.

B is the point with coordinates $(-1, 5)$

C is the point with coordinates $(2, 10)$

M is the midpoint of BC .

Find an equation of the line through the points A and M .

Give your answer in the form $py + qx = r$ where p , q and r are integers.

(5 marks)

Q35**PRACTICE**

Practice · Very Hard

CH 3 · TOPIC 4**6 marks**

12 $ABCD$ is a kite with $AB = AD$ and $CB = CD$

A is the point with coordinates $(-2, 10)$

B is the point with coordinates $\left(-\frac{27}{5}, 4\right)$

C is the point with coordinates $(4, -5)$

Work out the coordinates of D

(6 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

- | | | |
|-----------|--|--|
| 01 | Draw | Table of three points, ruler, the full range. |
| 02 | Read | y the subject: m in front of x , c alone. |
| 03 | Build | c from the axis, m from a big triangle; or two points. |
| 04 | $\parallel$ / $\perp$ | Same m ; flip and change sign; point gives c . |
| 05 | Shapes | Intercepts; right angles; solve lines together. |

BOTTOM LINE $y = mx + c \rightarrow m = \text{rise} \div \text{run} \rightarrow c = \text{where it crosses the } y\text{-axis.}$

Answers

Linear Graphs $y = mx + c$ for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • DRAW A LINE FROM ITS EQUATION

- | | | |
|-----------|---|---|
| Q1 | Dr Eslam • Table | $(-1, -3), (0, -1), (3, 5)$ joined |
| Q2 | Practice • Easy | line through $(-1, -5)$ and $(3, 7)$ |
| Q3 | Practice • Easy | (a) 1, 3, 5, 7, 9 (b) line through $(-2, 1)$ and $(2, 9)$ |
| Q4 | Practice • Easy | line through $(-2, -7)$ and $(4, 5)$ |
| Q5 | Practice • Medium | line through $(-2, -4)$ and $(2, 8)$ |
| Q6 | Edexcel 4MA1 • Oct/Nov 2020 • Paper 1H • Q3 | line through $(-2, 15)$ and $(3, -5)$ |

IDEA 02 • READ m AND c

- | | | |
|------------|-------------------|---|
| Q7 | Dr Eslam • Read | $y = -2x + 3$: $m = -2, (0, 3)$ |
| Q8 | Practice • Easy | $(0, 5)$ |
| Q9 | Practice • Easy | $(0, -\frac{2}{3})$ |
| Q10 | Practice • Medium | $y = 3x - \frac{5}{3}$: both $m = 3$ |
| Q11 | Practice • Medium | A and D ($m = 2$) |
| Q12 | Practice • Medium | $(8, 0)$ |
| Q13 | Practice • Hard | (a) $y = 3x + 4$: both $m = 3$ (b) line gives -8 ; $-6 > -8$: above |

IDEA 03 • EQUATION FROM A GRAPH OR TWO POINTS

- Q14** Dr Eslam • Two points $m = 2: y = 2x + 1$
- Q15** Practice • Easy $y = 2x + 1$
- Q16** Practice • Easy (a) rise 10, run 4 (b) $y = 2.5x - 2$
- Q17** Practice • Medium (a)(i) $-\frac{1}{4}$ (ii) $y = -\frac{1}{4}x + 5$ (b) $y = -\frac{1}{4}x - 2$
- Q18** Practice • Hard scale from $y = x - 1$: $L: y = 3 - 1.5x$
- Q19** Edexcel 4MA1 • June 2022 • Paper 1HR • Q10 $y = 1.5x - 1$
- Q20** Edexcel 4MA1 • November 2023 • Paper 2H • Q9 $y = 1.5x - 3$
- Q21** Edexcel 4MA1 • May 2024 • Paper 1HR • Q9 $y = -\frac{1}{2}x + 1$

IDEA 04 • PARALLEL AND PERPENDICULAR LINES

- Q22** Dr Eslam • Perpendicular $m = -\frac{1}{3}: y = -\frac{1}{3}x + 4$
- Q23** Practice • Easy e.g. $y = -4x + 1$
- Q24** Practice • Medium $y = 6x + 2$
- Q25** Practice • Hard $y = -\frac{3}{4}x - \frac{11}{4}$
- Q26** Practice • Very Hard $(28, 0)$
- Q27** Edexcel 4MA1 • June 2024 • Paper 2H • Q18 $\frac{5}{2}$
- Q28** Edexcel 4MA1 • November 2024 • Paper 2H • Q18 $y = \frac{1}{2}x + 7$
- Q29** Edexcel 4MA1 • June 2025 • Paper 2HR • Q10 $y = 6 - 5x$

IDEA 05 • LINES AND SHAPES

- Q30** Practice · Hard $y = -\frac{1}{2}x + \frac{3}{2}$
- Q31** Practice · Hard $(20, 0)$
- Q32** Practice · Very Hard $y = -2x + 21$
- Q33** Practice · Very Hard (a) $y = \frac{3}{2}x$ (b) $C(4, 6): \frac{52}{3}$
- Q34** Practice · Very Hard $5y + 3x = 39$
- Q35** Practice · Very Hard $D(4.6, 8)$